

Phase Separation of Positively Charged Polypeptide Solutions: Interplay between Electrostatics and Nonelectrostatic Specific Interactions

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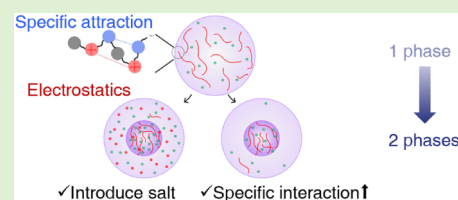
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ABSTRACT: The liquid–liquid phase separation (LLPS) of biomacromolecules governs various biological functions, such as ribosome biogenesis and cellular stress response. The understanding of the underlying driving forces, however, remains elusive. Here, we develop a simple mean-field theory to systematically examine how electrostatics and nonelectrostatic specific interactions influence the LLPS of positively charged polypeptide solutions. Our theory combines the generalized Debye–Hückel correlation, first-order perturbation for chain connectivity, and the sticker–spacer model for specific interactions. In salt-free systems, phase separation at a moderate charge fraction requires strong electrostatic correlation. For salty solutions without specific interactions, the miscibility gap exhibits a closed-loop feature on the polypeptide–salt concentration surface. Increasing the specific interaction strength or salt concentration can promote phase separation in otherwise homogeneous solutions, in good agreement with previous experiments and our molecular dynamics simulations. Our study reveals the intricate interplay among counterion entropy, excluded volume, electrostatic correlation, and specific interactions in regulating the LLPS of polypeptide solutions.



INTRODUCTION

The liquid–liquid phase separation (LLPS) of biomacromolecules, such as RNAs and proteins, plays a fundamental role in the formation of biomolecular condensates that regulate various cellular spatiotemporal functions, including cell differentiation, transcription regulation, and signal processing.^{1–4} For example, the P granules in *Caenorhabditis elegans* have been demonstrated to be liquid droplets formed by RNA and proteins via LLPS¹ and may contribute to the maintenance of the germline.⁵ Moreover, the aging of some liquid biomolecular condensates into solid amyloid fibrils is implicated with the neurodegenerative diseases such as Amyotrophic Lateral Sclerosis (ALS), Alzheimer's, and Parkinson's diseases.^{6–11} In the past 15 years, great efforts have been made to understand the formation mechanisms and functions of these biomolecular condensates, and significant progress has been achieved, we refer to the excellent reviews and perspectives for details.^{2,12–18}

The multivalent interactions among intrinsically disordered proteins (IDPs) or intrinsically disordered regions (IDRs) of proteins and RNAs are believed to be one of the main driving forces underlying the LLPS of biomolecular condensates.^{19–22} Actually, various experiments suggest that even the IDPs/IDRs themselves can phase separate under certain conditions.^{23,24} Different from the proteins that have well-defined tertiary native states, IDPs/IDRs usually consist of repeats of low-complexity domains^{25–28} that are rich in polar and charged amino acid residues^{29,30} and they usually adopt irregular

random coil conformations under physiological conditions. The sticker–spacer model, originally proposed in the context of associative polymer solutions^{31,32} is one of the most popular models to explain the LLPS underlying biomolecular condensates.^{33–37} In this model, repetitive domains on biomolecules act as stickers that can pair with each other, while other nonbinding regions are considered spacers. Based on this model, Pappu and coworkers developed Monte Carlo methods and mean-field theory to study the phase behaviors in solutions of the prion-like domain (PLD) of Fused in Sarcoma (FUS) proteins, as well as in binary mixtures of PLDs and hnRNPA1 proteins.^{38–45} In particular, they evaluated the relative importance of homotypic and heterotypic interactions in governing the phase behaviors of two-component biomolecules.^{45,46} Employing the same model, Ren et al. elucidated the dual role of adenosine triphosphate in regulating the phase behaviors of FUS solutions.⁴⁷

On the other hand, using the sticker–spacer model, Wingreen and coworkers performed coarse-grained molecular dynamics (CGMD) simulations to examine the effects of stoichiometry and valency on the bulk phase behaviors,

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interfacial properties, and chain dynamics of two-component condensates composed of IDP EPYC1 and holoenzyme Rubisco in the strong binding limit.^{48–52} Their main findings are further supported by their own dimer-gel theoretical calculations.^{53,54} Payne-Dwyer et al. proposed a semianalytical sticker–spacer theory to study the titration and dimerization in dilute mixtures of holoenzyme Rubisco and linker proteins.⁵⁵ Michels et al. generalized the sticker–spacer model to the two-component polymer blends in a single solvent and found that the delicate balance between specific binding and nonspecific interactions can lead to extremely rich phase behaviors.⁵⁶ Nandi et al. developed a simple lattice dimer–multimer model to account for the effects of affinity and valency on phase behaviors of multivalent proteins.⁵⁷ Moreover, using a similar lattice sticker–spacer model but without one-to-one pairing restriction, Panagiotopoulos et al. carried out a series of grand canonical Monte Carlo simulations to examine how the sequence affects the mesoscopic aggregation and macroscopic phase separation processes of biopolymers.^{58–63}

In spite of its popularity and success in modeling phase behaviors of biomacromolecules, the sticker–spacer model treats the associative interactions between sticker moieties in a highly coarse-grained way. At the molecular level, these associative interactions generally include the electrostatic interactions between oppositely charged amino acid residues, the π – π interactions between aromatic residues, the hydrophobic interactions between nonpolar residues, and other interactions such as cation– π and hydrogen bonding interactions. The sticker–spacer model simply treats these interactions as an effective interaction in a mixed manner and thus cannot evaluate the relative importance of different contributions. Therefore, it is unable to address many important questions, such as how the charge fraction, salt concentration, and nonelectrostatic specific interactions individually affect the general phase behaviors of IDPs. In fact, Rekhi et al. recently showed that the multiple interactions among all amino acid residues, instead of a few sticker residues, are responsible for the phase separations of IDPs.⁶⁴ On the other hand, several other theoretical models that explicitly treat the electrostatic interactions have been proposed to understand the role of electrostatics in the phase behaviors of IDPs and mixtures of IDPs and RNA. For example, by employing the Random Phase Approximation (RPA) theory, which explicitly considers the charge sequence effect in polyampholyte solutions, Chan and coworkers examined the phase behaviors of RNA helicase Ddx4 solutions at different temperatures and salt concentrations.^{65–67} Their pioneering work highlights the role of sequence-dependent electrostatic interactions. On the other hand, by performing field theoretical simulations, CGMD simulations, and RPA calculations, Fredrickson and coworkers systematically studied the effects of charge sequence, salt concentration, excluded volume interaction, and electrostatic correlation strength on the phase behaviors of polyampholyte solutions.^{68–71} By comparing their simulation results with the experimental observations on the LLPS of tau protein solutions and mixtures of tau and RNA, they further elucidated the distinct roles of electrostatics in governing the phase behaviors of different systems.⁷²

However, most of these electrostatic theories ignore the important role of other specific interactions, such as π – π and hydrogen bonding interactions, that commonly exist in biomolecular condensates and, therefore, cannot explain several experimental phenomena. For example, Agarwal et al.

found that the pathological truncation variant of the prion protein, comprising the unstructured *N*-terminal domain, undergoes LLPS only at finite salt concentrations, which is mainly driven by weak nonelectrostatic interactions.⁷³ Likewise, the addition of salt leads to LLPS in solutions of the PLD of hnRNPA1, in contrast to the salting-in behavior observed in solutions of full hnRNPA1, which consists of both the PLD and the folded domain.⁷⁴ Moreover, recent work by Guo et al.⁷⁵ demonstrated that an interplay between hydrogen bonding interactions and electrostatics between amino acid residues can lead to LLPS of polypeptides at finite salt concentrations. Actually, by treating the cation– π and π – π interactions via an effective Flory–Huggins interaction form, Lin et al. have already demonstrated the important role of these nonelectrostatic interactions under certain conditions.⁶⁵ However, this treatment neglects the one-to-one pairing feature between the sticker residues. To understand the above experimental findings, a unified theory that combines both electrostatic interactions and specific interactions, such as π – π and hydrogen bonding interactions, is required.

In this work, we develop a simple mean-field theory that explicitly incorporates electrostatic interactions and non-electrostatic specific interactions to examine the phase behaviors of polypeptide solutions. We treat the electrostatic correlation by incorporating the finite-sized Debye–Hückel theory⁷⁶ for disconnected ions and the first-order thermodynamic perturbation theory (TPT1)^{77–83} for chain connectivity. The specific interactions between aromatic residues or between hydrogen-bonding moieties are accounted for using the mean-field theory based on the sticker–spacer model.³² Using this theory, we systematically examine the interplay between electrostatics and specific interactions on the phase behaviors of polypeptide solutions. We find that, in salt-free cases, phase separation can be driven by either specific interactions or electrostatic correlations when their strengths are sufficiently strong. The addition of salt screens the electrostatic repulsion between charged amino acids and thus allows phase separation to take place at modest electrostatic correlation strength, consistent with previous experimental observations.^{75,84,85} Furthermore, we carry out coarse-grained molecular dynamics simulations to verify our theoretical predictions.

MODEL AND METHODS

Model. We consider a solution of n polypeptide chains, each having N amino acid residues, in a system of volume V . For simplicity, we assume that each peptide chain carries only positively charged residues, such as lysine and arginine; polar residues, such as glycine and asparagine; and aromatic residues, like tyrosine. This model is also motivated by recent experiments on polypeptide solutions in which the chain mainly consists of only one type of charged residue, polar residues, and aromatic residues. For example, Yu and coworkers showed that the phase behavior in solutions of polypeptides derived from the adhesive protein Mfp-5, GK-16*, is dominated by electrostatics and hydrogen bonding interactions.⁷⁵ On the other hand, the IDP Y145Stop, which carries around 10.4 positive charges, has been shown to phase separate with increasing salt concentration, mainly due to weak multivalent interactions.⁷³ It is worth mentioning that most IDPs are essentially polyampholytes that contain both positively and negatively charged amino acid residues. Our model presented here is a special case of such systems, containing only positively charged residues. Incorporation of both positively and negatively charged residues, as well as other hydrophobic residues, is straightforward but will increase the complexity of the system; we will reserve this consideration for the future. We denote the number of

charged residues and the number of residues capable of forming specific interactions (called sticker hereafter) on each chain as N_c and N_s , respectively, and the remaining $N_a = N - N_c - N_s$ groups are polar residues. Therefore, the fractions of charged residues and sticker residues are given by $\alpha = N_c/N$ and $\beta = N_s/N$, respectively. For simplicity, we assume that all charged groups are completely dissociated and further assume that each charged residue, counterion, and salt ion are all monovalent, with the counterion identical to the salt anion. Thus, the salt concentration is merely the cation concentration. Denoting the number of anions, cations, and solvent as n_- , n_+ , and n_0 , respectively, and assuming all species to have the same volume v , we have the volume fractions of polypeptide residues, cations, anions, and solvent as $\phi_p = nNv/V$, $\phi_+ = n_+v/V$, $\phi_- = n_-v/V$, and $\phi_0 = n_0v/V$, respectively. The excluded volume interaction among different species is modeled via the incompressibility constraint, i.e., $\phi_p + \phi_+ + \phi_- + \phi_0 = 1$. Further, the charge neutrality condition requires that $\alpha\phi_p + \phi_+ = \phi_-$.

Theoretical Framework. As alluded to earlier, here we ignore other contributions such as cation- π and hydrophobic interactions to highlight the roles of electrostatics and nonelectrostatic specific interactions such as hydrogen bonding and π - π interactions. In our system, the Helmholtz free energy density consists of the ideal part, the specific attraction part between sticker residues, and the electrostatic part, $f = f_{id} + f_a + f_{el}$. The dimensionless ideal free energy density accounts for the translational degrees of freedom of all species and is given by

$$f_{id} = \frac{\phi_p}{N} \ln \frac{\phi_p}{eN} + \phi_+ (\ln \phi_+ - 1) + \phi_- (\ln \phi_- - 1) + \phi_0 (\ln \phi_0 - 1) \quad (1)$$

Accurate modeling of the one-to-one specific interactions is usually a rather challenging task. Here, we adopt the simple mean-field theory developed by Semenov and Rubinstein³² to explain the formation of the nonelectrostatic specific bonds between sticker residues. Besides those studies on biomolecular condensates, as mentioned in the Introduction, this theory was also widely used in various polymeric systems. For example, Dormidontova generalized this theory to study the closed-loop phase diagrams of poly(ethylene oxide) (PEO) in water.⁸⁶ Ren et al. used this framework to study the effects of lithium salt on the phase behaviors of PEO/polystyrene blends.⁸⁷ Prusty et al. applied this theory to study the phase separations of polymer blends with hydrogen bonds.⁸⁸ Danielsen et al. generalized this theory to explore the phase separation and formation of reversible networks in solutions of two associative polymers.⁸⁹ Different from these studies, here we consider both the specific associative interaction and the electrostatics simultaneously, with a focus on the interplay between these two contributions.

Assuming that there are N_p sticker pairs formed in the solution, the number of residues participating in the pair formation is thus $2N_p$. Following the procedure by Rubinstein and Semenov,³² we derive the relevant Helmholtz free energy as

$$\frac{F_a}{k_B T} = -\ln \frac{(nN_s)!}{(nN_s - 2N_p)! 2^{N_p}} \left(\frac{v_p}{V} \right)^{N_p} \exp(-N_p \epsilon_s / k_B T) \quad (2)$$

where k_B is the Boltzmann constant and T is the absolute temperature. In eq 2, the first term in the logarithm accounts for the number of ways to form N_p pairs from a total of nN_s stickers, the second term denotes the loss of translational entropy due to the formation of N_p pairs, with v_p being the volume of each pair, and the last term represents the energy gain due to the formation of pairs, with ϵ_s characterizing the strength of specific attraction for each pair.

Defining $p = 2N_p/nN_s$ as the fraction of stickers that participate in the formation of sticker pairs and using Stirling's approximation, the dimensionless free energy density due to the specific attraction contribution reduces to

$$f_a \equiv \frac{F_a v}{k_B T V} = \frac{\phi_p N_s}{2N} \left[p \ln p + 2(1-p) \ln(1-p) - p \ln \frac{\phi_p N_s v_p}{eNv} + \frac{p \epsilon_s}{k_B T} \right] \quad (3)$$

Minimizing f_a with respect to p , we obtain the optimal p at a given parameter set satisfying

$$p = \frac{2 + A^{-1} - \sqrt{(2 + A^{-1})^2 - 4}}{2} \quad (4)$$

with $A \equiv (\phi_p N_s v_p / Nv) \exp(-\epsilon_s / k_B T)$; since v_p is on the same order of v , hereafter we take $v_p = v$ for simplicity. Substituting this optimal p into eq 3, we obtain the minimized f_a as

$$f_a = \frac{\phi_p N_s}{N} \left[\ln(1-p) + \frac{p}{2} \right] \quad (5)$$

We note that eqs 4 and 5 are the same as those developed by Semenov and Rubinstein.³²

Finally, we account for the electrostatic contribution due to the charged species using the generalized Debye-Hückel theory and the first-order thermodynamic perturbation (TPT1) theory. The Debye-Hückel theory accounts for the electrostatic correlation for disconnected ions, and the TPT1 theory is used to consider the extra free energy contribution due to the chain connectivity.⁸¹⁻⁸³ Therefore, the free energy density due to electrostatic correlation is composed of two contributions: one is for the disconnected ions, and the other arises from chain connectivity, i.e., $f_{el} = f_{DH} + f_{ch}$. For the first contribution, we use the celebrated generalized Debye-Hückel form⁷⁶

$$f_{DH} = -\frac{1}{4\pi} \left[\ln(1 + \kappa\sigma) - \kappa\sigma + \frac{(\kappa\sigma)^2}{2} \right] \quad (6)$$

where κ is the inverse of the Debye screening length given by $\kappa\sigma = \sqrt{4\pi(l_B/\sigma)(\phi_p\alpha + \phi_+ + \phi_-)}$. $l_B = e^2/4\pi k_B T \epsilon_0 \epsilon_r$ is the Bjerrum length, with ϵ_0 and ϵ_r being the permittivity of vacuum and the dielectric constant of the medium, respectively. For pure water at room temperature, $l_B \approx 0.74$ nm. $\sigma = v^{1/3}$ is roughly the diameter of each species. Different from the limiting law form $f_{DH} = -(\kappa\sigma)^3/12\pi$, eq 6 explicitly considers the excluded volume effect in the electrostatic correlation and thus has a wider application range. On the other hand, the electrostatic correlation due to the chain connectivity of polypeptides can be accounted for based on the TPT1 theory. This theory was first developed for the neutral tangentially connected hard sphere chain model⁷⁷⁻⁷⁹ and subsequently generalized to the polyelectrolyte hard sphere chain model.^{80,81} Using this theory, Zhang et al. studied the phase behaviors of fully charged polyelectrolyte solutions both in salt-free conditions and with salt.⁸² Later, Wang and co-workers generalized this theory to study the phase behaviors of partially charged polyelectrolyte solutions.^{90,91} Recently, Zhang and Wang further generalized this theory to account for the electrostatic correlation between adjacent charged monomers in the sense of the charged sequence.⁸³ In this theory, the leading-order contribution of the electrostatic correlation due to the chain connectivity for the partially charged system is considered. This theory has been demonstrated to correctly describe the main features of phase separation in polyelectrolyte solutions⁸² and polyelectrolyte complex coacervation.^{83,92} While the charge sequence effect can be directly considered in this theory, here, for simplicity, we assume the charges to be uniformly distributed along the chain backbone and thus the number of residues between two neighboring charged residues along the chain backbone as $N/(N_c - 1)$. Therefore, the electrostatic correlation free energy density due to chain connectivity is

$$f_{ch} = \frac{l_B \phi_p (N_c - 1) \exp[-\kappa \sigma (\sqrt{N/(N_c - 1)} - 1)]}{N \sigma (1 + \kappa \sigma) \sqrt{N/(N_c - 1)}} \quad (7)$$

Note that different treatments of electrostatics, such as RPA,^{65,93–96} dipolar attraction theory,^{97,98} scaling theory,⁹⁹ and renormalized Gaussian fluctuation theory,^{100,101} can also be applied to this system. We expect that the qualitative behaviors obtained using TPT1 remain similar.

In our theory, we follow the mean-field treatment by Semenov and Rubinstein to account for the sticker–sticker association, which is usually accurate when $N \gg 1$, $p \ll 1$, and ϕ_p is large. Moreover, there may exist some coupling terms between the sticker residues and charged residues in the free energy density. However, a theoretical treatment of these coupling effects is rather challenging and is still lacking in the literature. We thus ignore these coupling contributions in our work. The simplicity of our theory allows us to systematically examine the phase behaviors in a large parameter space. Furthermore, the overall agreement between our theoretical calculations and several experimental observations, as will be shown later, suggests that our theory captures the main factors determining the phase behaviors of our system.

In summary, the thermodynamics of our system is specified by a total of 7 parameters: the chain length N , the fraction of charged residues α , the fraction of sticker residues β , the Bjerrum length l_B that characterizes the strength of electrostatic correlation, the specific interaction parameter ε_s , the polymer concentration ϕ_p , and the salt concentration $\phi_s = \phi_+$. For clarity, we introduce the dimensionless variable $\tilde{\varepsilon}_s \equiv \varepsilon_s/k_B T$ to denote the strength of specific interactions. For brevity, we simply drop the tilde symbol in the following sections.

Construction of Phase Diagram. From the Helmholtz free energy density f , we calculate the exchange chemical potential of species i per residue basis as $\mu_i \equiv \partial f / \partial \phi_i|_{\phi_j \neq i}$, and the osmotic pressure as $\Pi = \mu_p \phi_p + \mu_+ \phi_+ + \mu_- \phi_- - [f - f(\phi_p = 0, \phi_+ = 0, \phi_- = 0)]$.¹⁰² When a polypeptide solution separates into a supernatant phase (represented by phase I) and a liquid coacervate phase (represented by phase II), the phase equilibrium condition requires the electrochemical potential and osmotic pressure of phases I and II to be equal. Denoting the volume fraction of species i in the supernatant and coacervate phases by ϕ_i^I and ϕ_i^{II} , respectively, we have

$$\begin{aligned} \mu_p(\phi_p^I, \phi_+^I, \phi_-^I) &= \mu_p(\phi_p^{II}, \phi_+^{II}, \phi_-^{II}) + \alpha \Psi_G \\ \mu_+(\phi_p^I, \phi_+^I, \phi_-^I) &= \mu_+(\phi_p^{II}, \phi_+^{II}, \phi_-^{II}) + \Psi_G \\ \mu_-(\phi_p^I, \phi_+^I, \phi_-^I) &= \mu_-(\phi_p^{II}, \phi_+^{II}, \phi_-^{II}) - \Psi_G \\ \Pi(\phi_p^I, \phi_+^I, \phi_-^I) &= \Pi(\phi_p^{II}, \phi_+^{II}, \phi_-^{II}) \\ \alpha \phi_p^I + \phi_+^I &= \phi_-^I \\ \alpha \phi_p^{II} + \phi_+^{II} &= \phi_-^{II}, \end{aligned} \quad (8)$$

where Ψ_G is the Galvani potential that represents the difference between the electrostatic potential in the coacervate phase and that in the supernatant phase.⁸² Physically, this potential arises from the charge neutrality constraint in each of the coexisting phases. In general, we have 6 equations with 7 variables Ψ_G , ϕ_i^I , and ϕ_i^{II} ($i = p, +, -$). Therefore, at a given parameter set, we need to fix one variable (such as ϕ_p^I) and calculate the other 6 variables by numerically solving the 6 equations using the Newton–Raphson method with appropriate initial guesses. By sweeping this fixed variable to all possible values, we can construct the complete binodal curves. In the absence of salt, the co-ion concentration is zero in both phases and thus $\phi_-^I = \phi_-^{II} = 0$; likewise, the second equation in eq 8 also vanishes. Therefore, we have 5 variables (Ψ_G , $\phi_p^{I/II}$, $\phi_+^{I/II}$) with 5 equations that can be uniquely solved.

Molecular Dynamics Simulation. To verify our theoretical results, we also perform coarse-grained Langevin dynamics simulations using the Large Atomic/Molecular Massively Parallel Simulation (LAMMPS) platform.¹⁰³ We treat the solvent as an implicit dielectric continuum and all other species as explicit beads interacting via the shifted Lennard–Jones purely repulsive potential $U_r(r) = 4\epsilon[(\sigma/r)^{12} - (\sigma/r)^6 + 1/4]$ when $r < r_c = 2^{1/6}\sigma$ and 0 otherwise. The chain connectivity of each polypeptide is modeled via the finitely extensible nonlinear elastic (FENE) potential given by $U_b(r) = -1/2KR_0 \ln[1 - ((r - \sigma)/R_0)^2]$ with $K = 30k_B T/\sigma^2$ and $R_0 = 1.5\sigma$. Following the works by Wingreen and co-workers^{48–50} and by Ryu et al.,¹⁰⁴ we model the specific interaction between two stickers as $U_a(r) = A[1 + \cos(\pi r/r_0)]$ when $r < r_0 = 1.5\sigma$ and 0 otherwise. Here, the parameter A is negative to denote an attractive interaction, and its absolute value reflects the strength of the specific interaction. Finally, the long-range electrostatic interaction between charged beads i and j is given by $U_{ei,j} = k_B T l_B q_i q_j / r_{ij}$ with q_i and q_j being the valence of beads i and j , respectively, and r_{ij} being their spatial distance. We employ the particle–particle/particle–mesh (PPPM) method to treat long-range electrostatic interactions with an accuracy of 1×10^{-5} . Assuming $\sigma = 0.4$ nm, we set $l_B = 1.785\sigma$ to represent the aqueous solutions at room temperature and $|q_i| = 1$ to denote the monovalent case for all species. In the following, the units of mass, energy, and length, as given by m , $k_B T$, and σ , respectively, are set to be 1. To save computational costs, we place 10 chains, each having 60 monomers, in a volume of $45\sigma \times 45\sigma \times 45\sigma$. Motivated by the experimental work by Yu and coworkers,⁷⁵ we mainly set $\alpha = 0.3$ and $\beta = 0.3$ in our simulations as a typical example to examine the roles of salt concentration and the strength of specific interaction.

The simulation evolves according to the Langevin dynamics with a time step of $\Delta t = 0.01$ in units of $\tau = \sqrt{m\sigma^2/k_B T} = 1$.^{105,106} To ensure the formation of a single coacervate droplet, we fix the center of mass of all polypeptide chains in the middle of the simulation box during the equilibration step and then increase $|A|$ gradually to the target values $|A_f|$. We choose $A_f = -2.5$ and -3.0 to denote different strengths of specific attraction. We run the simulation for 5×10^7 time steps for equilibration and then collect data every 100 steps over another 5×10^7 time steps. Specifically, we monitor the time evolution of the root-mean-square chain radius of gyration R_g to ensure the equilibration of our systems; once R_g does not change significantly and starts to fluctuate around some constant value, we regard that our simulation has reached equilibration.

RESULTS AND DISCUSSION

The thermodynamics of our polypeptide solution is determined by a total of 7 parameters: N , α , β , l_B , ε_s , ϕ_p , and ϕ_s . In the following, we first consider the salt-free case by setting $\phi_s = 0$ and examine the roles of electrostatics and specific interactions on the phase diagrams. We then study how the addition of a monovalent salt affects phase separation. While most of our results are reported for $N = 100$, we also briefly study the effect of N . Finally, we verify our main findings using coarse-grained molecular dynamics simulations. To compare our theoretical results with experiments, we set $\sigma = 4$ Å and convert the dimensionless volume fraction ϕ to the concentration ρ in units of M; simply put, $\rho = 1$ M corresponds to $\phi = 0.0385$. Using a hard sphere model with a single σ to represent all residues is definitely a crude approximation that may result in certain inaccuracies. The value of 4 Å is good for glycine residues but may underestimate the size of other residues, such as lysine. We thus do not make a quantitative comparison with experimental measurements. However, we expect that the dependences of phase behaviors on various parameters, such as charge fraction, specific interaction, and salt concentration, remain valid. Finally, it is worth mentioning that the polypeptide concentration in the

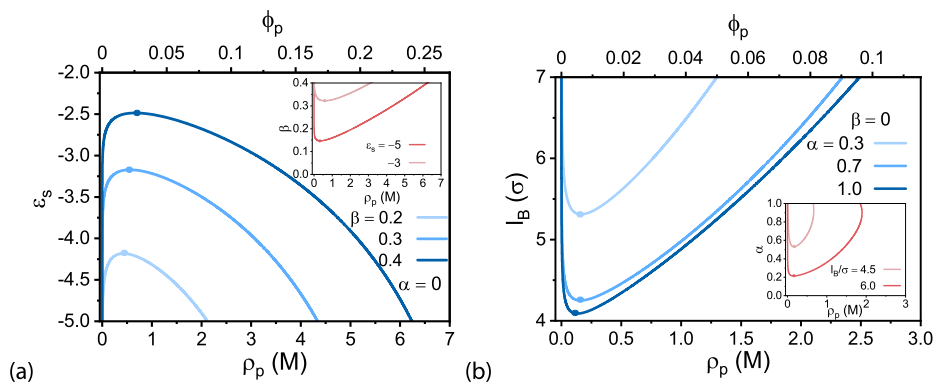


Figure 1. Binodal curves of the salt-free solutions. Panel (a) shows the binodal in the ρ_p – ε_s plane, and the inset shows the binodal in the ρ_p – β surface for systems with only sticker–sticker attraction. Panel (b) shows the binodal curves in the ρ_p – l_B surface, and the inset shows the binodal in the ρ_p – α surface for solutions with only electrostatic interactions. $N = 100$.

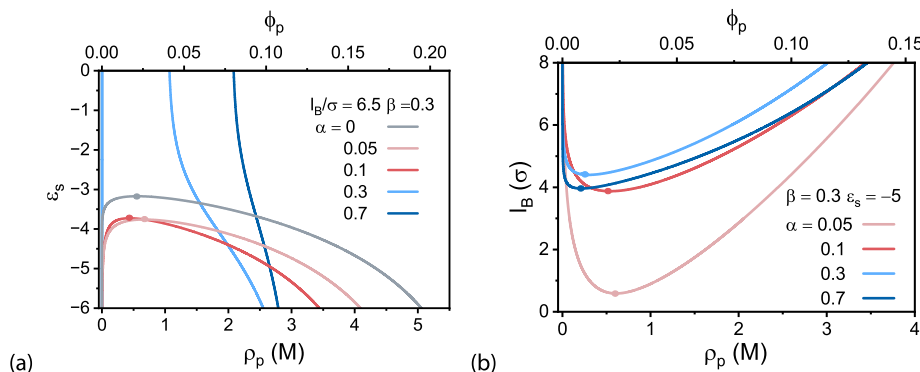


Figure 2. Binodal curves for the salt-free solutions with both electrostatic and nonelectrostatic specific interactions. Panel (a) shows the binodal in the ρ_p – ε_s surface while panel (b) shows the binodal in the ρ_p – l_B surface. $N = 100$.

entire article refers to the polypeptide concentration on a per-amino acid residue basis.

Salt-Free Cases. We first consider the salt-free case where the only small ion is counterion dissociated from the polypeptide backbone. At fixed N , the system properties are determined by 4 parameters: α and l_B control the electrostatics, while β and ε_s govern the nonelectrostatic specific interactions. Hereafter, we first examine their roles separately and then study their combined effects. Taking $N = 100$ as an example, in Figure 1a, we show the binodal curves in the ρ_p – ε_s plane at $\alpha = 0$ and various β . We choose $\beta = 0.2, 0.3, 0.4$ to cover the range of the fraction of hydrogen-bonding residues used in the experiment by Yu and co-workers.⁷⁵ The electrostatics play no role in this case. A solution is homogeneous above the curve but separates into a supernatant phase and a coexisting polypeptide-rich coacervate phase below it. It is apparent that the phase-separated region is wider for larger β . For example, phase separation will take place when $\varepsilon_s < -2.5$ at $\beta = 0.4$ and when $\varepsilon_s < -4.2$ at $\beta = 0.2$. Likewise, the inset of Figure 1a shows that the phase-separated region at fixed ε_s shrinks gradually with decreasing β , and the critical value of β below which a solution is homogeneous over all polypeptide concentrations is smaller for larger $|\varepsilon_s|$. Therefore, increasing either the specific interaction strength or the fraction of stickers can facilitate phase separation, which is consistent with previous results in the literature.^{32,75,107} Moreover, Figure 1a also shows that the volume fraction of polypeptide per-residue basis is less than 25%, and thus the volume fraction of water in the polypeptide-rich phase is more than 75%, suggesting that

the polypeptide-rich phase is a liquid coacervate phase instead of a solid precipitate phase within the parameter range we used.

On the other hand, in Figure 1b, we set $\beta = 0$ to examine the role of pure electrostatics on the phase behaviors. At fixed α , phase separation takes place only when l_B is larger than a critical value $l_{B,c}$ that increases with decreasing α . For example, $l_{B,c}/\sigma \approx 4.1$ at $\alpha = 1$ and 5.3 at $\alpha = 0.3$. The increase of $l_{B,c}$ with decreasing α is consistent with the previous observations by Wang and co-workers.^{90,91} In this case, phase separation is purely driven by the electrostatic correlation, and these results are consistent with previous studies.^{81,82,108–110} It is worth noting that $l_{B,c}$ in these cases is well above the value of l_B in pure water at room temperature (i.e., $l_B/\sigma \approx 1.785$). We thus conclude that a salt-free polypeptide solution in the absence of specific interactions is most likely homogeneous, and this is also consistent with experiments in aqueous solutions.^{75,111,112} However, we expect such a kind of phase separation to occur in organic solvents with a small dielectric constant. Indeed, the study by Lee et al. on polyelectrolyte complex coacervation suggests an enhanced electrostatic correlation in organic solvents with a low dielectric constant.¹¹³ Besides, the inset of Figure 1b shows that α should be sufficiently large to induce phase separation at fixed l_B , and the critical value of α increases with decreasing l_B , as expected.

We next examine the combined effects of electrostatics and nonelectrostatic specific interactions on the phase behavior of salt-free polypeptide solutions. In Figure 2a, we show the binodal curves in the ρ_p – ε_s plane at various α values and $\beta =$

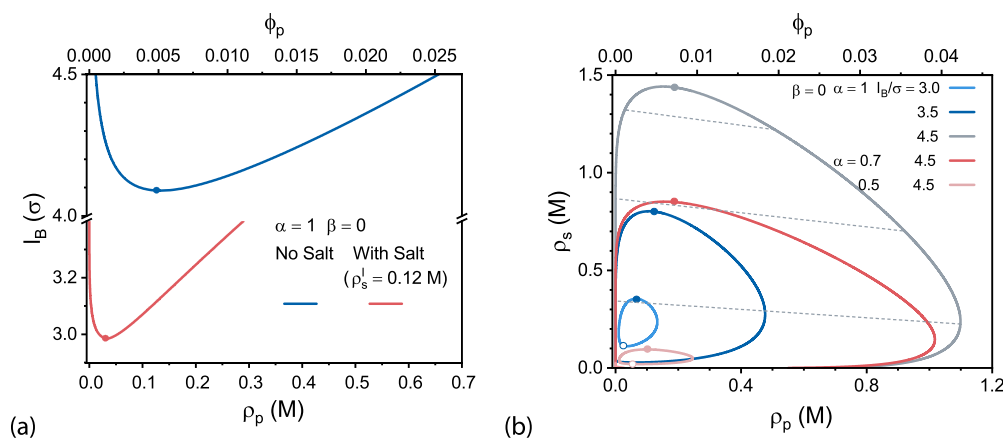


Figure 3. Binodal curves for polypeptide solutions without nonelectrostatic specific interactions in the presence of salt. Panel (a) shows the binodal curves in the ρ_p - l_B plane for the fully charged peptide without salt or with $\rho_s^I = 0.12$ M in the supernatant phase. Panel (b) shows the binodal curves in the ρ_p - ρ_s surface at various l_B and α . $N = 100$.

0.3; we choose this β value since it is reasonably close to the experimental setting used by Yu and coworkers.⁷⁵ We further use a relatively large value of $l_B/\sigma = 6.5$ since a strong electrostatic correlation is required to induce phase separation in the salt-free solutions. It is interesting to note that the phase-separated region first shrinks with increasing α , and a stronger specific interaction is required to induce phase separation when $\alpha < 0.1$. This is mainly because, at small α , the electrostatic correlation is rather weak, and the translational entropy of counterions, which increases with α , plays the dominant role in reducing the degree of phase separation.

With further increases in α , however, phase separation may take place even without specific interactions; see the two curves with $\alpha = 0.3$ and 0.7 in Figure 2a. The phase-separated window along the ρ_p axis, however, is much narrower than that at a small α . This feature reflects the rather distinct roles of electrostatics and specific interactions in phase separation. Different from the small α cases, where phase separation is dominated by specific interactions, the phase separation at large α is driven by both electrostatics and specific interactions. One interesting observation is that, at large $|\epsilon_s|$, the polypeptide concentration in the coacervate phase first decreases and then increases with increasing α . This feature is qualitatively consistent with the inset of Figure 1b. To understand this phenomenon, we note that increasing α brings two effects: 1. the concentration of counterions increases, and the accompanying translational entropy tends to inhibit phase separation; 2. the electrostatic correlation increases and thus it will promote phase separation. The competition between these two effects leads to a nonmonotonic variation of the polypeptide concentration in the coacervate phase along α .

To reveal the role of l_B on the phase behaviors, in Figure 2b we show the binodal curves in the ρ_p - l_B surface at $\beta = 0.3$ and various α . We set $\epsilon_s = -5$ to represent a reasonable specific interaction strength, which is sufficiently strong to drive phase separation in the absence of charge. When α is small, such as $\alpha = 0.05$, phase separation can take place when $l_B > l_{B,c} \approx 0.6\sigma$. With increasing α , $l_{B,c}$ increases significantly and then decreases slightly. For example, $l_{B,c}/\sigma \approx 4.0, 4.5$, and 4.0 at $\alpha = 0.1, 0.3$, and 0.7 , respectively. Again, this phenomenon results from the interplay among the specific interactions, the counterion translational entropy, and the electrostatic correlations. At small α , the specific interaction is the dominant factor in

driving phase separation. However, at large α , phase separation requires a sufficiently strong electrostatic correlation to win over the translational entropy of counterions, and thus it takes place only at relatively large l_B . Since $l_B/\sigma \approx 1.785$ in water at room temperature, our results suggest that a salt-free polypeptide solution with a reasonable number of stickers can phase separate only when its charge fraction is very small. In other words, a polypeptide solution with chains carrying a large number of charged residues tends to be homogeneous. This finding is also qualitatively consistent with the experimental observation by Yu and coworkers.⁷⁵

Effects of Salt. Salts have been demonstrated to play a significant role in determining the phase behaviors of various charged systems, such as single-component polyelectrolyte solutions, complex coacervates, and polyelectrolyte gels. In general, the valency and chemical identity of salt ions play an important role in determining the phase behaviors of polypeptide solutions, such as the mysterious Hofmeister effect, which was discovered more than a century ago but is still not fully understood today.^{114,115} In spite of these complexities, here we focus on simple monovalent salts, such as sodium chloride, and thus avoid the complex specific ion effects. Generally speaking, adding salt brings three effects: 1. increasing the translational entropy of salt co-ions and simultaneously increasing the translational entropy of counterions, 2. screening the electrostatic repulsion between the same charged residues, and 3. increasing the electrostatic correlation strength. While the first effect tends to suppress phase separation, the latter two facilitate phase separation. For solutions with purely specific interactions, i.e., $\alpha = 0$, the polypeptide chain does not carry any charge and thus salt scarcely influences the phase behaviors. In the following, we first examine how salts affect the phase behaviors of a polypeptide solution with only electrostatics and then study the phase behaviors for systems with both electrostatics and specific interactions.

With Only Electrostatics. To illustrate the effects of salt, in Figure 3a we present the binodal curves on the ρ_p - l_B surface for the fully charged polypeptide solution, with the salt concentration in the supernatant phase specified at $\rho_s^I = 0.12$ M. In comparison with the salt-free case (the blue curve), we find that the critical l_B decreases from $l_{B,c}/\sigma \approx 4.1$ to 3.0 , and the critical polypeptide concentration $\rho_{p,c}$ also decreases.

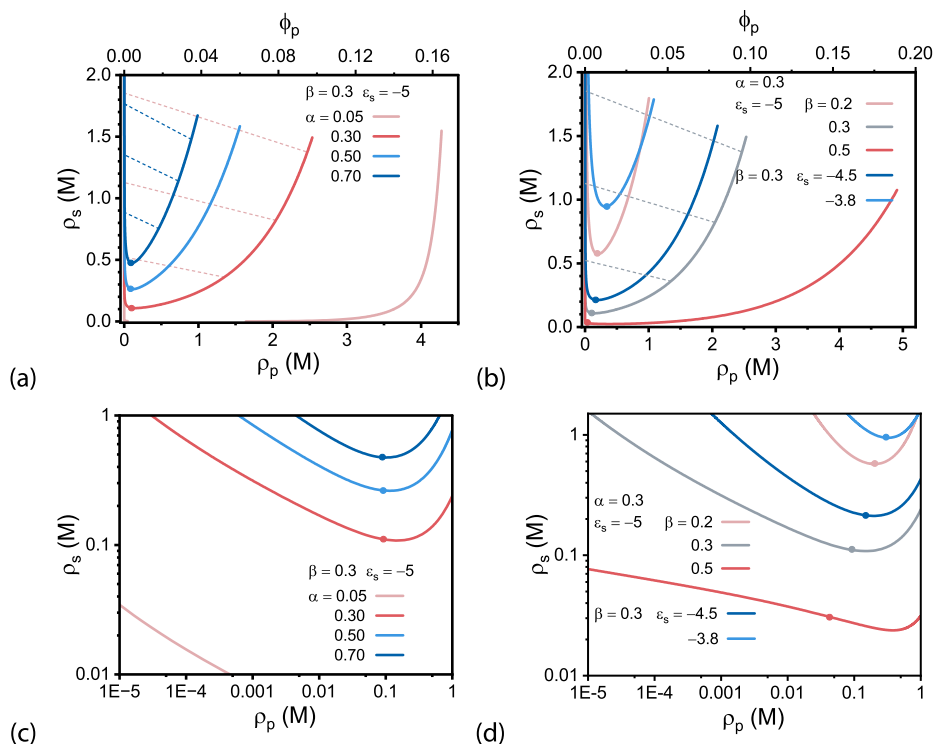


Figure 4. Binodal curves in the ρ_p – ρ_s surface for polypeptide aqueous solutions with both electrostatic and nonelectrostatic specific interactions in the presence of salt. Panel (a) shows the binodal at various α with $\beta = 0.3$, $\epsilon_s = -5$. Panel (b) shows the binodal at various ϵ_s and β at fixed $\alpha = 0.3$. Panels (c) and (d) are the log–log plots of panels (a) and (b), respectively, for a better presentation of the dilute branch in the phase diagram (i.e., the saturation concentration). $l_B/\sigma = 1.785$ and $N = 100$.

Therefore, the addition of salt expands the phase separation window. To further elucidate the effect of ρ_s in Figure 3b we present the binodal curves in the ρ_p – ρ_s phase diagrams at various l_B and α . Note that the vertical axis denotes the salt concentration. For the fully charged system with $\alpha = 1$, in Figure 1b we have already elucidated that a solution undergoes phase separation when $l_B/\sigma > 4.1$ and remains homogeneous otherwise. Therefore, at $l_B/\sigma = 4.5$, the miscibility gap first expands and then shrinks with the addition of salt. The tie lines all have negative slopes, indicating that the salt concentration in the supernatant phase is slightly higher than that in the coacervate phase. The upper critical point, at which the two coexisting phases merge, is located slightly to the right of the topmost point. When l_B/σ is decreased to lower values, such as 3.5 and 3.0, we observe closed-loop binodal curves with both an upper critical point and a lower critical point. In other words, while an initial salt-free solution is homogeneous, the addition of salt may lead to phase separation. The solution may become homogeneous again when the salt concentration is sufficiently high. These salting-out and salting-in behaviors are consistent with previous theoretical work by Zhang et al.,⁸² which used a different version of liquid-state theory for fully charged polyelectrolyte solutions.

On the other hand, with decreasing α , the miscibility gap shrinks gradually and vanishes eventually. It is interesting to note that the binodal curve at $\alpha = 0.5$ is of the closed-loop type, and phase separation is inhibited when the salt concentration is either too high or too low; this feature is rather different from the phase diagrams at $\alpha = 0.7$ and 1.0, where phase separation may take place even in the absence of salt, i.e., $\rho_s = 0$. These results are fully consistent with the observed critical value of $\alpha \approx 0.53$ at $l_B/\sigma = 4.5$ shown in the

inset of Figure 1b. In other words, the critical charge fraction in the salty solution may be lower than that in salt-free solutions. This is again due to the competition between the translational entropy of small ions and the electrostatic correlation.

With these results, we conclude that the addition of a monovalent salt may lead to a decrease in the critical values of l_B and charge fraction α . However, the critical l_B above which phase separation takes place is still large in comparison to that of water at room temperature. While it is accessible in organic solvents with lower dielectric constants, phase separation of polypeptides in aqueous solution is usually hard to achieve with pure electrostatics.

With Both Electrostatics and Specific Interactions. Having demonstrated the role of salt in the system with only electrostatics, we next examine how the combination of electrostatics and nonelectrostatic specific interactions affects the phase behavior of polypeptide solutions. Here, we fix $l_B/\sigma = 1.785$ to represent the aqueous solution at room temperature. In Figure 4a, we show the binodal curves in the ρ_p – ρ_s planes at various α , with $\beta = 0.3$ and $\epsilon_s = -5$. At each parameter set, phase separation occurs when the salt concentration ρ_s is above the critical salt concentration $\rho_{s,c}$. When the charge fraction is minute (such as $\alpha = 0.05$), although a solution can phase separate without salt, the addition of salt leads to a significant expansion of the phase-separated window. Increasing α leads to a narrowing of the miscibility gap and an increase of the critical salt concentration. When the charge fraction is large (such as $\alpha = 0.3, 0.5$, and 0.7), the solution is homogeneous in the absence of salt, but a certain amount of salt will induce phase separation. Physically, this can be understood as the electrostatic repulsion between polypeptides being screened at large salt concentrations,

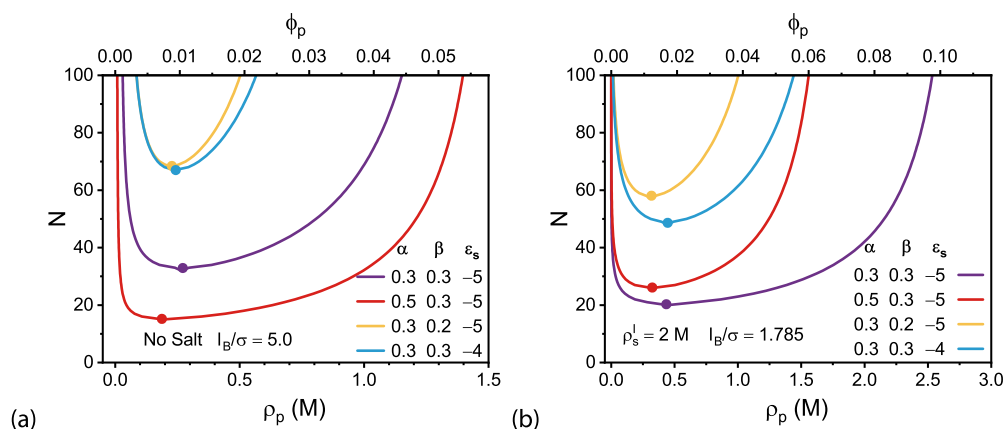


Figure 5. Binodal curves in ρ_p - N plane for (a) the salt-free solution and (b) the solution with fixed salt concentration in the supernatant phase $\rho_s^I = 2$ M at various α , β , and ϵ_s . The solid circles denote the respective critical points.

allowing the specific interaction to drive the phase separation. The higher the charge fraction, the larger the amount of salt required for this screening. Moreover, for all cases, the salt concentration in the supernatant phase is higher than that in the coacervate phase, and the difference in salt concentration between these two phases is more significant for larger α . Finally, different from the solution with only electrostatics, salting-out behaviors are observed only in this system.

Previous experiments found that the saturation concentration of the IDP/IDR ranges from several to hundreds of μ M, and the salt concentration required for phase separation is on the order of hundreds of mM.^{38,75,85} To facilitate a better comparison with experimental findings, in Figure 4c we replot the binodal curves on a log-log scale. We find that the saturation concentration of the polypeptide, characterized by the dilute branch of the binodal, decreases with increasing salt concentration and covers the relevant experimental values. Furthermore, Babinchak et al.⁸⁵ found that with increasing pH, the net charge of the LCD region of the transactive response DNA-binding protein of 43 kDa (TDP-43) decreases, and phase separation takes place. On the other hand, increasing salt concentration leads to either the appearance or enhancement of phase separation. They regarded that the addition of salt screens the charge repulsion, and then the hydrophobic interaction between nonpolar residues drives the phase separation. Likewise, Agarwal et al.⁷³ and Ferrand et al.¹¹⁶ also observed similar salting-out phenomena in prion protein variant solutions driven by weak multivalent interactions and in histidine-rich beak protein 2 (HBP-2) solutions driven by π - π interactions, respectively. All of these experimental results are qualitatively consistent with our theoretical calculations.

Now let us turn to examine how the fraction of stickers β and the specific interaction strength ϵ_s influence phase behaviors. Taking $l_B/\sigma = 1.785$ and $\alpha = 0.3$ as an example, in Figure 4b we show the binodal curves in the ρ_p - ρ_s surface at various combinations of β and ϵ_s . For a typical specific interaction $\epsilon_s = -5$, the phase-separated region expands with increasing β , as expected. Similarly, decreasing $|\epsilon_s|$ at fixed β leads to a shrinkage of the miscibility gap, and phase separation may vanish completely when $|\epsilon_s|$ is too small. In a word, a reasonable amount of stickers with sufficiently strong interaction is required to induce phase separation for charged polypeptide solutions at room temperature. In Figure 4d, we further replot this phase diagram in a log-log plot for better visualization of the dilute branch of the binodal. The saturation

concentration decreases with an increase in either the salt concentration or the fraction of sticker residues. Yu and coworkers found that the replacement of tyrosine with dopa in hexadecapeptide can increase hydrogen bonding strength and further lead to phase separation at finite salt concentrations.⁷⁵ Our results, shown in Figure 4b and d are also qualitatively consistent with these experimental observations.

Chain Length Dependence. All of the above calculations are carried out with the chain length ($N = 100$). In fact, chain length also plays a key role in governing phase behaviors. It is generally acknowledged that the synthesis of polypeptides with a large number of amino acid residues is a highly complex process, and thus understanding the chain length dependence of phase behaviors will provide valuable suggestions for designing polypeptides with appropriate chain lengths in experiments. Therefore, in this section, we will examine how chain length affects the phase behaviors of peptide solutions. In Figure 5a, we first present the binodal curves in the ρ_p - N diagram for the salt-free solutions at various α , β , and ϵ_s . We choose $l_B/\sigma = 5.0$ since phase separation is hard to access at lower l_B for salt-free solutions under typical parameter sets. In general, the miscibility gap shrinks with decreasing N and phase separation no longer occurs when N is too short. For example, when $\alpha = 0.3$, $\beta = 0.3$, and $\epsilon_s = -5$, phase separation cannot take place when $N < 32$. Moreover, the critical N below which a solution is always homogeneous, decreases with increasing values of β , α , and ϵ_s .

We next examined the chain length effect for solutions with salts. Yu and coworkers examined the salt effect on the phase behaviors of polypeptide solutions to mimic mussel adhesion in seawater by increasing the salt concentration up to 2M.⁷⁵

Here, by taking the case with $\rho_s^I = 2$ M as an example, in Figure 5b we show the binodal curves in the ρ_p - N diagram at various α , β , and ϵ_s . Here, we choose $l_B/\sigma = 1.785$ not only because it is experimentally accessible but also to further elucidate the effect of different l_B . In general, the binodal curves exhibit similar behaviors to those of the salt-free cases. One exception is the effect of α on the phase boundary. While increasing α leads to an expansion of the miscibility gap for the salt-free case at $l_B/\sigma = 5$, it shrinks the phase separation window for the salty solutions at $l_B/\sigma = 1.785$. Physically, this is due to the complicated role of electrostatics at different l_B . At a small l_B , the electrostatic repulsion between the charged monomers dominates over the electrostatic correlation and

thus suppresses phase separation. On the other hand, at large l_B , the electrostatic correlation becomes more important, and it provides a further driving force for phase separation besides the specific interaction, and thus leads to an enhancement of phase separation. We also calculate the binodal curves at other extra salt concentrations, and the general features of the binodal curves are similar to those in Figure 5.

Molecular Dynamics Simulations. To verify our theoretical predictions, we also perform Langevin dynamics simulations for polypeptide solutions, either in the salt-free case or with finite salt concentrations. We choose the sequence shown in Figure 6a where different types of amino acid

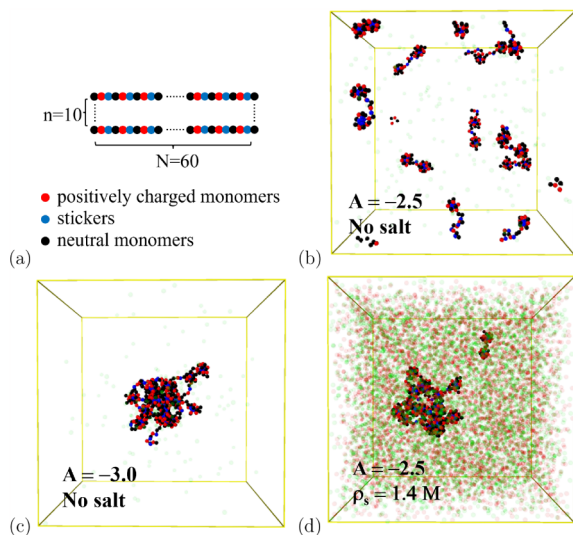


Figure 6. (a) Schematics of polypeptide chains in coarse-grained MD simulation with both charged and sticker residues being evenly distributed. (b) Snapshot of MD simulation for a salt-free solution with $A = -2.5$. Panels (c) and (d) are the simulated snapshots for the salt-free solution with $A = -3.0$ and the salty solution with $A = -2.5$ and $\rho_s = 1.4\text{ M}$, respectively.

residues are distributed in a relatively uniform way. This sequence is closer to our theory, where both charged and sticker residues are uniformly distributed along the chain backbone. By taking the case with $N = 60$ and $\rho_p = 0.17\text{ M}$ in a volume of $(45\sigma)^3$ as an example, in Figure 6 we show the typical snapshots at several representative values of specific interaction strength A and salt concentration ρ_s . Here, we set $\alpha = \beta = 0.3$ and $l_B/\sigma = 1.785$. Figure 6b shows that, in the salt-free

solution, all chains are dispersed in the system when $A = -2.5$. This reflects the fact that the sticker–sticker attraction here is weak in comparison with the electrostatic repulsion between charged residues on the chain. With increasing $|A|$ to 3.0, the sticker–sticker attraction is sufficiently strong to overcome the repulsion between charged residues, and all chains are aggregated into a large cluster, as shown in Figure 6c. To quantitatively characterize the structure of this cluster, in Figure 7a we present the densities of polypeptide and counterions, denoted by ρ_p and ρ_- , respectively, along the radial direction r from the center of mass of the cluster. We note that ρ_p exhibits a peak at finite r and approaches 0 when r is large. The inset of Figure 7a shows that the density of the charged residue ρ_{p+} and that of the sticker residue ρ_{st} exhibit similar behaviors. On the other hand, the counterion density ρ_- decreases roughly monotonically with r . To characterize the charged state of the cluster, the magenta curve in Figure 7a shows the net charge density $\rho_e/e = \rho_{p+} - \rho_-$ along r , with e being the elementary charge; it is positive inside the cluster and approaches 0 when r is large. The repulsion arising from this finite charge density inside the cluster may explain the nonmonotonic variation of ρ_p along r inside the cluster. With further increasing $|A|$, the cluster becomes denser, and the polypeptide concentration inside the cluster increases. The cluster may enter a solid precipitate state with a kinetically frozen structure, which was examined using all-atomic MD simulations.^{117,118} However, due to both the challenge in accurate theoretical modeling and the lack of reliable experimental data for the solid precipitate phase, we do not pursue the investigation of this phase in the current work.

As mentioned earlier, the addition of salt can screen the electrostatic repulsion between charged residues and result in phase separation. To verify this finding, in Figure 6d we present the snapshot for the system with $A = -2.5$ at $\rho_s = 1.4\text{ M}$. Despite the fact that the corresponding salt-free solution is dispersed (see Figure 6b), most of the chains are aggregated into a cluster when $\rho_s = 1.4\text{ M}$. We next examine the structure of this cluster in Figure 7b. We see that the peptide concentration also shows a peak at finite r and approaches 0 at large r . As expected, inside the cluster, the anion concentration is higher, while the cation concentration is lower than their respective bulk concentrations. An interesting observation is that the charge density, calculated via $\rho_e/e = \rho_{p+} + \rho_+ - \rho_-$, exhibits an oscillating electric double-layer structure. Specifically, ρ_e achieves a maximum around $r \approx 6.7\sigma$ and undergoes a decaying oscillation to 0 with further

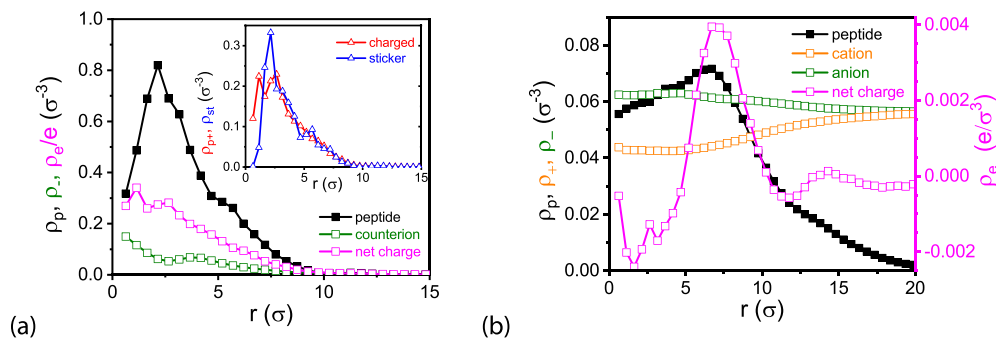


Figure 7. MD results of radial density distributions of polypeptide, small ions, and net charge for (a) the salt-free solution with $A = -3.0$ and (b) the salty solution with $A = -2.5$ and $\rho_s = 1.4\text{ M}$. The inset of panel (a) shows the radial density distributions for charged residues and sticker residues. $\beta = 0.3$, $\alpha = 0.3$, and $l_B = 1.785\sigma$.

increasing r . On the other hand, ρ_e deep inside the cluster is overall smaller than 0, indicating the occurrence of charge inversion. While the data here are somewhat noisy due to the insufficient sampling of the simulation, we expect that the qualitative physics has been captured. A more systematic and quantitative examination of these behaviors will be pursued in the future. Moreover, in comparison with the cluster shown in Figure 6c, the cluster formed at finite salt concentration is looser, with a smaller overall peptide concentration and a larger size.

We also make some quantitative comparisons of the coexisting concentrations obtained from the mean-field calculation and MD simulation. As one example, the MD simulations predict that, at $A = 2.5$ and $\rho_s = 1.4$ M, the polymer concentrations in the supernatant and in the coacervate phase are $\rho_p^I \approx 0.162$ M and $\rho_p^{II} \approx 0.733$ M, respectively, and the salt concentrations (i.e., cation concentration) in these two phases are $\rho_s^I \approx 1.42$ M and $\rho_s^{II} \approx 1.32$ M, respectively. As a comparison, the mean-field theory at $\varepsilon_s = -3.8$ yields the polymer concentrations in the two coexisting phases as $\rho_p^I \approx 0.087$ M and $\rho_p^{II} \approx 0.788$ M, and the salt concentrations (i.e., cation concentration) in these two phases as $\rho_s^I \approx 1.42$ M and $\rho_s^{II} \approx 1.30$ M. We note that the coexisting concentrations obtained from these two methods are in reasonable agreement. Moreover, both methods predict that the salt concentration in the supernatant phase is higher than that in the coacervate phase. However, it is worth mentioning that such comparisons are only qualitative since different model systems are used in these two methods. Moreover, the finite-size effect in the simulations may also lead to some quantitative differences. Indeed, the work by Zhang and Wang suggests that, in symmetric polyelectrolyte complex coacervation, a smaller cluster has a higher internal polymer concentration due to the Laplace pressure effect.¹¹⁹ Nevertheless, these comparisons suggest that our theoretical formulation indeed captures the main physical ingredients governing the phase behaviors of polypeptide solutions.

CONCLUSIONS AND OUTLOOK

In this work, we have developed a simple mean-field theory to investigate the liquid–liquid phase separation of polypeptide solutions with both electrostatic and nonelectrostatic specific interactions. We consider a polypeptide solution in which the chain mainly consists of positively charged residues, polar residues, and aromatic residues, and examine how the fraction of charged groups and their interaction strength, denoted by α and l_B , respectively, and the fraction of associative groups and their interaction strength, denoted by β and ε_s , respectively, affect the phase behaviors for solutions without salt and in the presence of monovalent salt. For the salt-free solutions, either the electrostatics or the specific interaction can solely drive phase separation when the relevant interaction is strong enough, and the related critical strength depends on the fraction of the respective groups. The combination of these two interactions leads to a nonmonotonic dependence of the phase boundary on the charge fraction at large electrostatic strengths. While the increase of charge fraction α initially shrinks the miscibility gap due to the increased role of the translational entropy of counterions, further increasing α expands the phase-separated window due to the stronger electrostatic correlation at larger α .

For solutions without specific interactions, the addition of salt decreases the critical value of l_B , and the binodal curve in the ρ_p – ρ_s plane adopts a closed-loop behavior at moderate l_B below the critical value in the salt-free case. This indicates both salting-out and salting-in phenomena when salt is added to a polypeptide solution. On the other hand, for solutions with moderate specific interactions, the addition of salt into a homogeneous polypeptide solution may lead to LLPS and the miscibility gap expands monotonically with increasing salt concentration. For all cases, the salt concentration in the supernatant phase is higher than that in the coacervate phase. Investigation of the chain length effect suggests that the chain must be long enough to induce phase separation. For aqueous solutions at room temperature, the critical chain length decreases with increasing β , ε_s , and decreasing α . We also employed CGMD simulation to verify our main theoretical predictions. Specifically, we find that phase separation can be induced by increasing the specific interaction strength between sticker residues or the salt concentration, in reasonable agreement with our theoretical calculations. We rationalize our results in terms of the interplay between the multifaceted aspects of electrostatics, including electrostatic correlation and intrachain repulsion, the translational entropy of counterions, and the specific interactions. The above findings are qualitatively consistent with systematic experimental observations in mussel foot protein solutions by Yu and coworkers,⁷⁵ in solutions of TDP-43 LCD regions by Babinchak et al.,⁸⁵ and in HBP-2 solutions by Ferrand et al.¹¹⁶ Specifically, these studies found the significant role of electrostatic and nonelectrostatic interactions in driving LLPS. Our theoretical results here not only demonstrate the intricate interplay between various interactions in governing the formation of biomolecular condensates but also provide additional insight into the mechanisms underlying phase separation in biomolecular systems.

Finally, we close this paper by making the following remarks. In this work, we have assumed that all charged residues carry a positive charge, and both charged and sticker residues are uniformly distributed along the chain backbone. Therefore, our main results reported here are restricted to understanding the phase behaviors of solutions of IDPs/IDRs carrying only positively charged residues, such as lysine and arginine, in addition to aromatic residues, such as tyrosine, and polar residues, such as glycine. In many systems, however, polypeptides are polyampholytes that carry both positive and negative residues with a well-defined sequence. The amounts of these oppositely charged residues, as well as their spatial arrangement along the chain backbone, are expected to have a profound impact on the phase behaviors of polypeptide solutions.¹²⁰ Meanwhile, different types of salt may result in rather different effects on the phase behaviors.¹²¹ Moreover, a variety of systems contain multicomponent biopolymers, such as RNAs and IDPs/IDRs. The phase behaviors in these systems are not only governed by homotypic and heterotypic sticker–sticker interactions but are also strongly affected by the electrostatic interactions. Previous studies have already reported rather complex phase behaviors in these systems, including microscopic and macroscopic phase separation,^{58,63} gelation,^{36,75} and multiphase separation.^{96,122} Last but not least, the phase separation kinetics of polypeptide solutions, via either nucleation and growth or the spinodal decomposition mechanism, is another important topic. Recent work by Shimobayashi et al. showed that the classical nucleation

mechanism still holds in the formation of some biomolecular condensates,¹²³ but a general picture of the effects of charge and specific interactions on phase separation kinetics remains lacking. A complete and unified understanding of these issues should be pursued in the future. Our study here represents an important step toward this direction.

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Notes

The authors declare no competing financial interest.

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